Climate Informatics 2026 (2026), xx: 1–2
doi:10.1017/xxxx

CAMBRIDGE UNIVERSITY PRESS

CLIMATE INFORMATICS 2026
EXTENDED ABSTRACT

Towards a mass-conservative global sea ice emulator that generalizes across climates

William Gregory[1*], Mitchell Bushuk[2], James Duncan[3], Elynn Wu[3], Adam Subel[4], Spencer Clark[2,3], Oliver Watt-Meyer[3], Alistair Adcroft[1], Chris Bretherton[3] and Laure Zanna[4]

[1]Atmospheric and Oceanic Sciences Program, Princeton University, NJ, USA
[2]Geophysical Fluid Dynamics Laboratory, NOAA, Princeton, NJ, USA
[3]Allen Institute for Artificial Intelligence (Ai2), Seattle, USA
[4]Courant Institute School of Mathematics, Computing, and Data Science, New York University, New York, NY, USA
*Corresponding author. Email: wg4031@princeton.edu

Received xx xxx xxxx

Keywords: climate informatics, sea ice, climate emulation

1. Introduction and objectives

We introduce FloeNet, a data-driven emulator architecture trained on the Geophysical Fluid Dynamics Laboratory (GFDL) global sea ice model, SIS2. FloeNet is an auto-regressive graph neural network (GNN) which marks a step forward in sea ice emulation as the first model to dynamically evolve the state of sea ice and snow-on-sea-ice by mass and area budget decompositions. Specifically, FloeNet receives mechanical and thermodynamic forcing inputs from the atmosphere and ocean, and predicts ice and snow mass tendencies due to growth, melt, and advection. This yields a mass-conservative and interpretable model.

Sea ice is an integral component of the coupled climate system, regulating Earth's surface temperature and playing a significant role in high-latitude climate feedbacks. Sea ice is also a barometer for climate change. It is therefore crucial that data-driven sea ice models show an accurate response to different climate forcings. To this end, we show preliminary results of stand-alone sea ice emulation using prescribed atmosphere and ocean forcing under different climates. We train FloeNet using data from a reanalysis-forced ice-ocean simulation (hereafter, OM4) over the period 1969–2005. We then evaluate three experiments: 1) a 54-year rollout using OM4 forcing over the period 1969–2022, 2) a 140-year rollout using pre-industrial control (piControl) forcing, 3) a 140-year rollout using 1% CO_2 forcing.

2. Methods and data

FloeNet is based on the GraphCast model [1], with some modifications. We choose a graph-based model as they allow us to flexibly set the degree of non-locality within the model. Specifically, we avoid global connections as we do not expect physically-meaningful timestep-to-timestep relationships between, for example, Arctic and Antarctic sea ice. We do however expect some amount of non-locality to be important, given that elastic stresses due to floe interactions propagate rapidly across basin-scales. We therefore configure FloeNet to have a receptive field of $\sim$1500 km.

Our OM4 simulation is run at 0.25° global horizontal resolution, which we coarsen to a regular 1°-resolution latitude-longitude grid before training. The atmospheric forcing data for driving FloeNet

© The Authors(s) 2020. Published by Cambridge University Press. This is an Open Access article, distributed under the terms of the Creative Commons Attribution licence (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted re-use, distribution, and reproduction in any medium, provided the original work is properly cited.

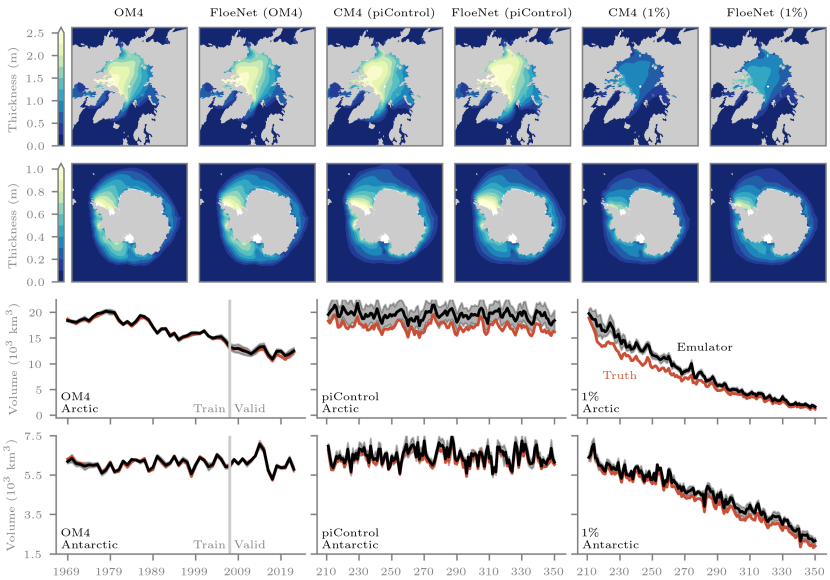

Figure 1. *Annual-mean sea ice thickness and volume under present-day (OM4), pre-industrial, and 1% CO_2 climate forcing. The grey vertical line in the OM4 time series panels show the FloeNet training period. Grey shading shows 3σ from 3 training runs initialized with different random seeds.*

are 6-hourly averages of downward shortwave and longwave radiation, sensible and latent heat fluxes, zonal and meridional wind stress over sea ice, and snowfall. The ocean forcing data are 5-day averages of basal heat flux, sea-surface height, and zonal and meridional ice-ocean stress. Prognostic variables are 6-hour snapshots of ice concentration, ice mass, snow mass, and ice-surface skin temperature, as well as 6-hour averages of growth, melt, and transport mass and area fluxes for ice and snow (9 total), ice-to-ocean salt flux, and top and bottom energy fluxes. The piControl and 1% forcing data are taken from experiments which were run using GFDL's CMIP6-generation fully coupled climate model, CM4.

3. Preliminary results and expected outcomes

Fig. 1 shows the annual-mean sea ice thickness (spatial maps) and sea ice volume (time series) from OM4/CM4 numerical simulations and FloeNet. We find that FloeNet is able to capture the spatial pattern and magnitude of the sea ice thickness climatology in both hemispheres, under all three forcings. In terms of sea ice volume, FloeNet shows clearly physical responses to changes in forcing, even reaching ice-free conditions in summer under the 1% forcing (not seen in Fig 1); FloeNet also accurately captures the correct magnitude of inter-annual variability.

We conclude that FloeNet is a fast and accurate global sea ice emulator, taking 4.75 hours to generate a 140-year simulation on 1 GPU. We expect that FloeNet will substantially improve the representation of atmosphere-ice-ocean interactions in existing climate emulators, the subject of ongoing work.

Acknowledgments. This project is supported by Schmidt Sciences, under the M^2LInES project. Ai2 is supported by the estate of Paul G. Allen.

References

[1] Lam, R., Sanchez-Gonzalez, A., Willson, M., Wirnsberger, P., Fortunato, M., Alet, F., Ravuri, S., Ewalds, T., Eaton-Rosen, Z., Hu, W. and Merose, A. (2023). Learning skillful medium-range global weather forecasting. *Science*, **382(6677)**, pp.1416-1421.